

WEEK 6 PROJECT SUMMARY

HealthConnect — Appointment Attendance & No-Show Analysis

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Track: Data Analytics

Reporting Period: Week 6

Collaborator: [Dorsilla Kemunto] Data Science

Prepared for: AnalystLab Africa — HealthConnect Week 6 Assignment

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1. What I Planned to Accomplish

Building on my Week 5 report, my Week 6 plan was to:

- **Validate all Week 5 KPIs** against the full dataset to confirm accuracy
- **Re-verify the reminder effectiveness calculation** the Week 5 gap (1.3 pp) needed checking against the corrected no-reminder attendance rate
- **Deepen the reminder channel analysis** using **attendance rate** as a complementary lens to Week 5's no-show rate
- **Segment no-shows by Distance × Age Group** to identify high-risk patient profiles
- **Identify the highest-risk combinations** of distance and booking lead time
- **Analyze attendance by Booking Lead Time × Appointment Type**
- **Integrate with the Data Science track** to cross-validate my analytical findings against a predictive model
- **Test Week 5's channel recommendation** Week 5 recommended WhatsApp as primary; Week 6 would test whether that holds under deeper analysis

2. What I Completed

All planned Week 6 work was completed:

Deliverable	Status
KPI re-validation against full dataset	✓ Complete
Reminder effectiveness correction (1.3 pp → 4.9 pp)	✓ Complete
Reminder channel analysis using attendance rate	✓ Complete
Distance × Age Group heatmap	✓ Complete
Highest-risk combinations (Distance × Lead Time)	✓ Complete
Lead Time × Appointment Type matrix	✓ Complete
Week 6 Power BI dashboard with new visuals	✓ Complete
Cross-track integration with Data Science	✓ Complete

Deliverable	Status
Cross-Track Validation Card added to dashboard	✅ Complete
Week 6 report + cross-track integration report + README update	✅ Complete

3. What I Improved from Week 5

Area	Week 5	Week 6
Metric lens	No-show rate only	Both no-show rate and attendance rate
Reminder effectiveness	Reported 1.3 pp gap	Corrected to 4.9 pp (47.6% with reminder vs 42.7% without)
Reminder channel recommendation	WhatsApp as primary	SMS primary; Email complementary supported by attendance-rate ranking (SMS 49.6% → Email 46.5% → WhatsApp 44.6% → None 42.7%)

Area	Week 5	Week 6
Segmentation	One dimension at a time	Distance × Age, Distance × Lead Time, Lead Time × Type
Risk identification	None	Top 6 highest-risk combinations identified
Validation	Internal only	Externally validated against a Random Forest model
Dashboard	2 tabs, baseline KPIs	Full diagnostic dashboard

Key improvement: Week 5's reminder effectiveness calculation (1.3 pp) was **miscalculated**. Correcting it in Week 6 revealed reminders are nearly four times more effective than originally reported (**4.9 pp gap**, not 1.3 pp). This correction materially changes the Week 5 narrative — reminders *do* work, just modestly.

4. What I Integrated

A. Analytical integration — combined four dimensions previously analyzed in isolation:

- Distance × Age Group
- Distance × Booking Lead Time
- Booking Lead Time × Appointment Type

- Reminder Channel × Volume × Attendance Rate

B. Cross-track integration — exchanged outputs with the Data Science track:

- Shared my Week 5 & Week 6 dashboards and segmentation dimensions
 - Received their Random Forest feature importance, target/feature setup, and error distribution
 - Performed **cross-method triangulation** between manual analysis and ML ranking
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5. Which Track(s) I Collaborated With

Data Science track — with [Dorsilla Kemunto].

This was a **two-way collaboration** with artifacts exchanged in both directions, not just communication. The collaboration produced a validated output, a data-quality correction, and a decision change (detailed in Sections 7 and 10).

6. What Was Exchanged

I provided (Analytics → Data Science):

- Week 5 dashboard — KPI validation, no-show rate by reminder channel
- Week 6 dashboard — attendance rate by channel, distance × age, risk combinations, lead time × type
- Week 6 segmentation dimensions — distance buckets, lead-time buckets, reminder channel, appointment type
- Manual analytical findings identifying **booking lead time** and **distance to clinic** as top drivers

I received (Data Science → Analytics):

- Target/feature setup — target = no_show; 4 features in Week 5 ML-ready set

- Random Forest feature importance ranking
- Error distribution (TN, TP, FN, FP)
- Model class and accuracy (~53%)

7. What Changed as a Result

Four concrete changes — not just information exchange:

#	Change	Type
1	Analytical findings independently validated. Manual segmentation and Random Forest feature importance both ranked booking lead time and distance to clinic as the top two drivers.	Validated output
2	A model defect was identified and reported. numeric_Unnamed: 0 appeared as the #3 most important feature an index leak from the Week 5 dataset, not a real predictor.	Improvement
3	The DS feature roadmap was redirected. I provided Week 6 segmentation dimensions as candidate features for the next model iteration, expanding beyond the current 4-feature limit.	Integration

#	Change	Type
4	The model's success metric was redefined. I recommended shifting from overall accuracy (~53%) to recall on the no-show class , because False Negatives are the costliest error in a scheduling context (FN = 242, nearly equal to TP = 243).	Decision

8. Key Findings / Development Outcomes

Analytical findings (Week 6):

- **Overall attendance rate:** 46.3% — unchanged from Week 5, confirming validation
- **Reminder effectiveness corrected:** gap is **4.9 pp** (47.6% with reminder vs 42.7% without), not the 1.3 pp reported in Week 5
- **SMS has the highest attendance rate (49.6%)** and highest volume (2,000 appointments)
- **Email is second at 46.5%** — a strong complementary channel
- **WhatsApp is third at 44.6%**, below Email despite higher volume — Week 5's WhatsApp-first recommendation is refined to **SMS-primary / Email-complementary**
- **No reminder produces the lowest attendance (42.7%)** — confirming reminders do help, just modestly
- **Follow-up + 1–2 Months = 29.6% attendance** — lowest segment in the dataset
- **General Consultation + Short Term = 72.2% attendance** — highest segment
- **Very Far distance elevates no-show across all age groups (53–83%)**
- **Highest single risk combination: Very Far + 2–4 Weeks = 83.3% no-show**

Cross-track development outcomes:

- Analytics and Data Science **converged on the same top drivers** — booking lead time and distance via two independent methods
 - Model accuracy (~53%) is insufficient for deployment and requires feature enrichment + class-imbalance handling
 - Model error-cost analysis reframed the objective from accuracy → recall
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9. Major Challenges

- **Reminder effectiveness miscalculation inherited from Week 5.** The Week 5 without-reminder attendance rate was recorded as 46.3% (same as overall attendance), which produced a misleadingly small 1.3 pp gap. Re-verifying against the correct no-reminder figure (42.7%) required revisiting the Week 5 calculation and correcting the headline narrative around reminder effectiveness.
- **Metric ambiguity between no-show rate and attendance rate.** The two metrics rank channels differently. Week 5's no-show-rate lens ranked WhatsApp best; Week 6's attendance-rate lens ranked SMS best. Resolved by clarifying the two metrics measure different outcomes (they do not sum to 100% because cancellations and reschedules exist between them) and by ranking channels on attendance rate: **SMS → Email → WhatsApp → None.**
- **Data quality issue in the ML-ready dataset.** Unnamed: 0 was present as a feature, introducing an index leak into the model.
- **Limited feature set** in the Week 5 ML-ready dataset (only 4 features, one invalid) constrained model performance.
- **Class imbalance in the model.** False Negatives (242) nearly equaled True Positives (243), reflecting poor recall on the minority no-show class.
- **Cross-track coordination** required aligning metric definitions between the two tracks so the validation exercise was meaningful.

10. Important Decisions

- **Decided to correct, not hide, the Week 5 reminder effectiveness figure.** Reporting 4.9 pp (not 1.3 pp) changes the narrative reminders *do* work and is the honest result of re-running the calculation.
 - **Decided to analyze channels using both no-show rate and attendance rate**, rather than replacing one with the other.
 - **Decided to refine the channel recommendation from WhatsApp-primary to SMS-primary with Email as complementary**, based on the attendance-rate ranking: SMS (49.6%) → Email (46.5%) → WhatsApp (44.6%) → None (42.7%). This is a data-driven refinement, not a reversal, Week 5's no-show lens and Week 6's attendance lens measure different outcomes.
 - **Decided to escalate the Unnamed: 0 index leak** to the Data Science track.
 - **Decided to recommend recall over accuracy** as the model's primary success metric.
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11. Remaining Issues

- **Why WhatsApp underperforms Email on attendance despite higher volume** needs investigation — is it a demographic effect, cancellation/reschedule pattern, or channel engagement issue?
- **Model accuracy remains ~53%** — requires feature enrichment and class-imbalance handling before it is deployment-ready.
- **Feature enrichment not yet applied** — the Week 6 dimensions I provided still need to be tested in the next model iteration.
- **Interventions not yet designed** — the highest-risk combinations have been identified but not yet addressed operationally.

- **Metric source of truth** — a project-level decision is still needed on whether no-show rate or attendance rate should be the primary KPI for channel evaluation going forward.
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12. My Contribution to the Overall HealthConnect Project

My contribution to HealthConnect has been the **analytical lens that turns raw appointment data into actionable, validated insight**.

Specifically:

- I built the **diagnostic segmentation layer** — identifying not just *that* patients miss appointments, but *which* patients, *under what conditions*, and *through which channels*.
- I **corrected a material miscalculation from Week 5**. The reminder effectiveness gap was reported as 1.3 pp; Week 6 established it as **4.9 pp** (47.6% vs 42.7%). This changes the project's understanding of whether reminders work — they do, just modestly.
- I identified the **top risk combinations** the project will target for intervention (Very Far + 2–4 Weeks = 83.3% no-show; Follow-up + 1–2 Months = 29.6% attendance).
- I **validated my findings against the Data Science model**, converging two independent methods on the same top drivers — strengthening the project's overall confidence in its recommendations.
- I **flagged model defects and reframed the evaluation metric**, contributing to the Data Science track's next iteration.
- I produced a **dashboard and written reports** that translate analysis into decisions HealthConnect can act on — **SMS-primary, Email-complementary reminders**, targeting Very Far patients, and staged reminders for long-lead bookings.

In short: the Data Science track brings the predictive capability, and I bring the **interpretive, validated business insight** that makes the model useful in practice.

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